THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A magnet can separate: 1

(a) iron filings from sulphur **(b)** sugar from water
(c) sand from salt **(d)** oil from water

2. A solution contains 20 mL of ethanol in 80 mL of solution. The volume percentage is: 1

(a) 80 per cent **(b)** 25 per cent **(c)** 20 per cent **(d)** 10 per cent

3. Crystallisation differs from simple evaporation because it: 1

(a) works for gases
(b) needs no heat
(c) boils to dryness
(d) gives pure crystals and does not evaporate to dryness

4. The solubility of most solids in water: 1

(a) decreases with temperature **(b)** does not change
(c) increases with temperature **(d)** becomes zero

5. The particles of a true solution are: 1

(a) smaller than 1 nm and do not scatter light
(b) visible to the naked eye
(c) settling down
(d) larger than 1000 nm

6. Water and kerosene (immiscible) can be separated by: 1

(a) a separating funnel **(b)** evaporation
(c) winnowing **(d)** sublimation

7. Aeration of drinking water and removing suspended impurities is done by: 1

(a) sublimation **(b)** sedimentation and filtration
(c) chromatography **(d)** winnowing

8. Which of these shows the Tyndall effect? 1

(a) sugar solution **(b)** milk **(c)** vinegar **(d)** salt solution

9. The Greek words in chromatography mean: 1

(a) light and sound **(b)** colour and writing
(c) heat and cold **(d)** water and heat

10. Air can be separated into its gases by: 1

(a) sublimation **(b)** filtration

(c) fractional distillation of liquid air **(d)** decantation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is sublimation? Name two substances that sublime. 2

12. State two differences between a true solution and a suspension. 2

13. What is an alloy? Give one example and its composition. 2

14. Why is the separation of a mixture into its components important? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Differentiate between a solution, a suspension and a colloid on size, stability and the Tyndall effect. 3

16. List three properties of colloids. 3

17. Explain why the oil and water do not mix but alcohol and water do. 3

OR

A solution contains 36 g of salt in 100 g of water at 20 degrees Celsius. Find the mass by mass percentage. 3

18. Explain how muddy water is purified in a water treatment plant. 3

OR

Explain the use of the Tyndall effect in identifying colloids with an experiment. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain concentration of solutions: mass/mass, volume/volume and mass/volume percentages with one numerical example for each. 5

OR

Explain the preparation of a colloid such as starch solution and how it differs from a true solution by observation. 5

20. Explain how solubility depends on temperature and nature of the solute and solvent. Use the solubility of salt and potassium nitrate as examples. 5

OR

Describe how suspensions and colloids are separated, including filtration, coagulation, centrifugation and dialysis. 5

END OF QUESTION PAPER — Total: 40 Marks